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PALAEOMAGNETIC DATA, LATE PRECAMBRIAN MAGNETOSTRATIGRAPHY AND TECTONIC EVOLUTION OF EASTERN CHINA

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ABSTRACT

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In this paper middle and upper Proterozoic (1900—1600 Ma) APW paths are presented for North and South China based on palaeomagnetic results obtained from the Jixian, Yangzi Gorges, Shennongjia and Fuzhou areas in eastern China. Two or three "loops" are present; the second of these has an age of about 1000—800 Ma and is probably roughly equivalent to the Sveconorwegian and Grenville loops. APW paths, loops, reversed polarity zones and pole positions, combined with geological data, permit definition of several markers that are useful in regional correlation. These form the basis for a proposed stratigraphic subdivision of the middle and upper Proterozoic in China. A tentative palaeogeographic map has been drawn for the North China and Yangzi Blocks for the period from 1700 Ma to 600 Ma.

INTRODUCTION

There are significant geological disparities between North and South China. Research on the evolutionary relationships between these areas is of great importance in establishing an integrated tectonic and palaeogeographic picture. This paper is mainly concerned with the relevance of palaeomagnetic data to these problems. In recent years the authors and Li Pu (Zhang Wenzhi and Li Pu, 1980; Zhang Huimin et al., 1983) have made preliminary systematic studies of late Precambrian rocks ranging in age from 1900 Ma to 600 Ma. These studies have been carried out at Jixian, Yangzi Gorges, Shennongjia and in the Fuzhou area where there is good exposure and the structure is relatively simple. These sections are suitable for measurement of remanent magnetization in the rocks. They are also the stratotype and parastratotype sections for middle and upper Proterozoic rocks in China (Chen Jinbiao et al., 1980; Wang, Yulen et al., 1980; Zhao Zigiang et al., 1980; Chen Jinbiao et al., 1981).

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TABLE I

Summary results from formations at sections in the Yangzi Gorges, Shennongjia, Fuzhou and Jixian areas

Area	Stratigraphic unit		S	N	Direction		A_{95}°	Pole position		ψ_p°	Ages Ma	Refer.
	Fm., Gr.	Ind.			D°	I°		λ°	ψ°			
Yangzi Gorg.	Dengying	Z_{dn}	5	56	39	19	14	223	48	10	670	Zhang H.-M. and Zhang W.-Z., 1983
	Doushantuo	Z_d	12	135	50	18	11	215	38	9	700	
	Liantuo	Z_l	13	69	40	47	13	194	55	28	750	
Shennongjia	Shuijintuo	C_s	3	27	30	34	14	218	59	18	C	
	Nantuo	Z_n	7	47	41	23	19	188	47	12		
	Datangpo	Z_{dt}	3	20	43	53	55	185	53	28	728	This paper
	Gucheng	Z_g	2	7	193	−30	49	251	−70	−16		
	Macaoyan	P_m	4	36	104	52	12	166	6	32	843	
	Dayanpin	P_d	4	38	62	42	40	191	35	24		
	Songziyan	P_{sn}	3	19	59	15	15	120	81	40		
	Shicaohe	P_s	5	26	238	0	18	200	−28	0		
	Taizi	P_t	4	37	258	51	30	53	7	32	1332	
	Kuangshishan	P_k	2	11	315	15	16	1	42	8		
	Dawokeng	P_{dw}	1	11	348	19	11	319	66	10		
	Luangshigou	P_l	3	26	289	59	25	50	34	40		
Fuzhou	Ganjingzi	Z_{gn}	14		30	69	8	169	66	52		Gao R.-F. et al., 1983
	Nanguanling	Z_{nn}	12		35	49	9	219	60	30		
	Changlingzi	Z_c	19		63	72	14	166	47	57	723	
	Qiaotou	Z_q	4		259	-64	9	357	−34	46		
	Nanfen	Q_n	14		31	72	6	162	64	56	787	
	Diaoyutai	Q_d	2	22	28	39	11	239	61	22	818	This paper
	Yongning	Q_y	1	12	62	57	12	194	43	37		
	Henan Xingji	C_x	2	12	17	58	6	178	75	39	C	
Jixian	Jingeryu	Q_j	2	16	12	75	14	132	67	62		

TABLE I (*continued*)

Area	Stratigraphic unit		S	N	Direction		A_{95}°	Pole position		ψ_p°	Ages Ma	Refer.
	Fm., Gr.	Ind.			D°	I°		λ°	ψ°			
	Chang-Longshan	Q_c	3	24	26	35	18	243	60	19	855	Zhang W.-Z. and Li P., 1980
	Xiamaling	Q_x	2	29	71	40	39	200	29	23	956	
	Teiling	J_t	4	34	193	−36	30	84	−67	−20	1046	
	Hongshuizhuang	J_h	3	19	157	− 5	20	322	47	2	1220	
	Wumishan	J_w	9	81	247	2	17	26	4	1		
	Yangzhuang	J_y	6	48	263	55	12	66	17	35		
	Gaoyuzhuang	N_g	4	21	271	13	5	32	5	7	1550 1434	
	Dahongyu	N_d	7	59	288	35	20	32	36	20		
	Tuanshanzi	C_t	3	23	284	22	8	27	18	11	1763	
	Chuanlinggou	C_{ch}	2	16	300	35	14	24	35	19	1848	
	Changzhougou	C_c	3	28	318	45	24	20	53	27		

Note: Fm. — Formation; Gr. — Group; Ind. — Index of Stratigraphic unit; S — Number of sampling sites; N — The number of samples; A_{95} — Radius of circle of confidence; ψ_p - Paleolatitude.

SAMPLING AND METHODS

To obtain formation-mean palaeomagnetic data, we sampled at least three sites in each formation and mainly sampled specimens of strongly magnetized rocks such as red sandstones, purple or purplish-red argillaceous dolostones and marls. Natural remanent magnetization of all samples was measured with a JR4 spinner magnetometer having a noise level of about 3.10^{-8} emu. Alternating field demagnetization was applied to all specimens at about 200—600 Oe. Some specimens were subjected to thermal demagnetization at 300 to 500°C after AF cleaning. The results show that red silty fine grained sandstones, purple, red argillaceous dolostones etc. have comparatively stable directions of magnetization; they possibly represent stable magnetization acquired during diagenesis or shortly afterwards (Table I).

RESULTS

Based on palaeomagnetic results from the Jixian, Yangzi Gorges, Shennongjia and Fuzhou areas, the apparent polar wander (APW) paths are similar, moving from Africa and the North Atlantic Ocean to the Northern

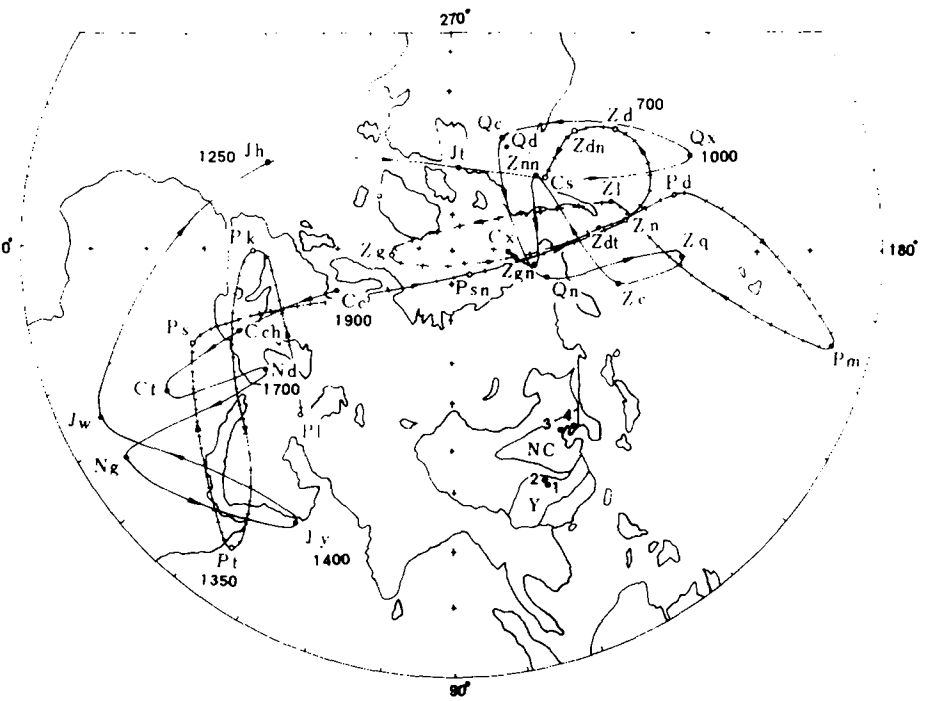


Fig. 1. Paths of apparent polar wander relative to North and South China for middle and upper Proterozoic.
Note: The locations of sampled sites: 1. Yangzi Gorges; 2. Shennongjia; 3. Jixian; and 4. Fuzhou. NC — North China Block, Y — Yangzi Block.
Symbols P_t, J_y, etc. are same as Table I.
1400, 1350, etc. are isotopic ages in Ma.

Pacific (Fig. 1). It is possible that the two blocks had similar drift-histories during the middle and upper Proterozoic. Thus it may be possible to carry out magnetostratigraphic correlations between the two blocks.

Two or three "loops" have been recognized in the paths. One is the Yangzhuang—Taizi loop, with an age of about 1500—1300 Ma. Another is the Xiamaling—Macaoyan loop, which is between about 1000 and 800 Ma. A possible third one is the Qiaotou—Gucheng loop, spanning the period from about 750—700 Ma. The second loop can be approximately correlated with the Sveconorwegian loop (1000—800 Ma) in northwestern Europe and the Grenville loop (1000—800 Ma) in North America (Piper, 1980; Pesonen and Neuvonen, 1981). These may provide data for intercontinental correlation.

Figure 2 shows changes of palaeolatitude in the mid- and upper Proterozoic of eastern China. In this period both blocks were mainly situated in low and intermediate latitudes between the equator and 40°. The North China block reached 62° N in the Jingeryu stage of the Qingbaikou Period. During that period the Fuzhou area was also above 50° N for part of the time. Palaeoclimatic and palaeoenvironmental indicators in sedimentary rocks at that time are consistent with the inferred palaeolatitudes of the various stages. Some carbonate units that are rich in stromatolites were located in tropical to subtropical latitudes, e.g., the Gaoyuzhuang, Wumishan and Yemahe-Shicaohe Formations. Most of the dark, fine clastics and mudstones appear to have been deposited in middle latitudes, e.g., the Taizi, Songziyan, Xiamaling and Datangpo Formations. Some manganiferous, iron-

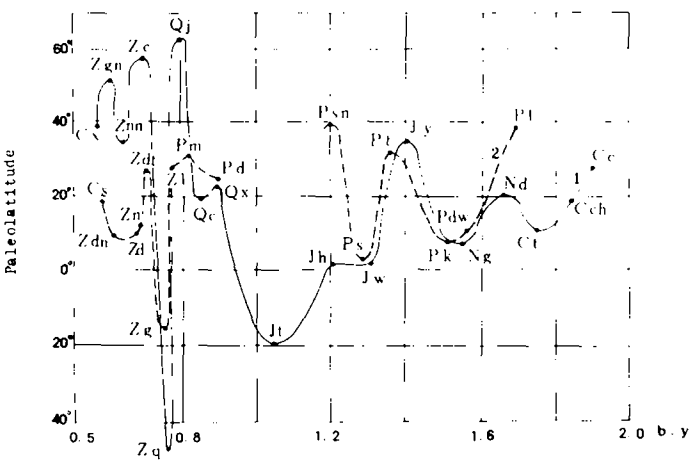


Fig. 2. Palaeolatitude diagram for Yangzi Gorges, Shennongjia, Jixian and Fuzhou Areas.
1. Shennongjia and Yangzi Gorges sections;
2. Jixian and Fuzhou sections.
Symbols P_t, J_y, etc. are same as Table I.

rich and phosphatic strata formed in regions of low palaeolatitude. The Nantuo and Gucheng tillites of the Sinian Period appear to have been deposited in low palaeolatitudes, consistent with data from some co-eval tillites on other continents (McWilliams and McElhinny, 1980).

Recognition of Precambrian polarity reversals is quite complex. Reversed polarity zones were established here according to the recognized concepts and techniques. They have been found in the Qiaotou and Gucheng Formations of Sinian age in the Fuzhou and Shennongjia regions, respectively. Reversed polarity was also detected in the Jixian section in the Teiling Formation which is about 1200—1000 Ma old. The Qiatou and Gucheng reversed polarity zones are approximately the same age and may serve as an important marker for magnetostratigraphic correlation between North and South China in the Sinian Period.

IMPLICATIONS OF RESULTS

Directions of magnetization and polarity characteristics provide important data relating to the sequential history of the Earth's magnetic field and the relative positions of crustal blocks. The record of magnetization in the Precambrian can potentially provide data comparable to those provided by fossil metazoa in the Phanerozoic. In fact the study of remanent magnetization is complex because of superimposed influences of diatsrophism, magmatism and other thermal events throughout the long intervals of geologic time. In general, the older the rocks, the more complex the history is likely to be.

Studies of APW paths, especially "loops", pole positions, palaeolatitudes and reversed polarity zones, taken in conjunction with geochronological palaeontological and geological data (Chen Jinbiao et al., 1980; Wang Yulen et al., 1980; Zhao Ziqiang et al., 1980), suggest that a few magnetostratigraphic marker beds might be present. Such markers may permit detailed correlations over wide areas.

Several such marker beds are shown in the magnetostratigraphic correlation scheme (Table II):

(1) Yangzhuang—Taizi magnetic marker bed. The evidence is: (a) they are located at the first "loop" (Fig. 1); (b) they have similar pole positions and palaeolatitudes. The pole position of the Yangzhuang Formation is $\lambda = 60^\circ$ E, $\psi = 17^\circ$ N and $\psi_p = 35^\circ$ N. The data for the Taizi Formation are $\lambda = 53^\circ$E, $\psi = 7^\circ$N and $\psi_p = 32^\circ$N; (c) according to U—Pb age determinations the age of the Taizi Formation is 1332 Ma and the age of the Yangzhuang Formation is bracketed between about 1350 and 1450 Ma according to data from the overlying and underlying units; and (d) both formations occur at the base of a sedimentary sequence and are overlain by a huge thickness of stromatolitic dolostones. Thus the Yangzhuang and Taizi Formations could serve as one of the major markers for correlation of the Jixian and Shennongjia sections;

(2) Xiamaling—Dayanpin magneto-biologic marker. The main lines of support for this marker are: (a) both units are situated in the second "loop" (Fig. 1); (b) their pole positions and palaeolatitudes are similar. The pole position for the Xiamaling Formation is $\lambda = 200°$ E, $\psi = 29°$ N and $\psi_p = 23°$ N. For the Dayanpin Formation the corresponding data are $\lambda = 191°$ E, $\psi = 35°$ N and $\psi_p = 24°$ N; (c) stromatolite assembages are similar in carbonates of the two units; and (d) both formations were deposited during a fairly tectonically active period;

(3) Changlongshan—Diaoyutai marker. Data are: (a) both units are situated, palaeomagnetically, in the terminal part of the Xiamaling "loop" (Fig. 1); (b) they have similar pole positions and palaeolatitudes, $\lambda = 243°$ E and 239° E, $\psi = 60°$ N and 61° N, $\psi_p = 19°$ N and 22° N, respectively; (c) lithology and stratigraphic context are identical; (d) Chuaria-type megafossils have been found in both units; and (e) the Changlongshan Formation has a glauconite K—Ar age of 855 Ma, the Diaoyutai Formation has a K—Ar age of 818 Ma from the same mineral, and

(4) Qiaotou—Gucheng reversed polarity marker. Relevant data include the following: (a) both units have reversed polarity or antiparallel reversals (Ray and Park, 1969; Elston and Bressles, 1980); (b) the palaeomagnetic pole positions fall on the third "loop". The ages of the overlying units are approximately the same. The Datangpo Formation has a Rb—Sr isochron age of about 728 Ma; the lower part of the Changlingzi Formation has a Rb—Sr isochron age of c. 723 Ma; and (d) worm traces occur in the upper part of the Changlingzi Formation as do metazoa, Chuaria and worm traces in the Doushantuo and Dengying Formations.

These markers provide a framework for correlations such as that shown in Table II, for the middle and upper Proterozoic of China. They also provide a means of correlating between the Jixian and Yangzi Gorges type sections in North China and areas in South China. The whole Jixian section, the upper part of the Nanfen Formation of the Fuzhou section and the Sinian section of the Yangzi Gorges represent the complete succession of the middle and upper Proterozoic of China.

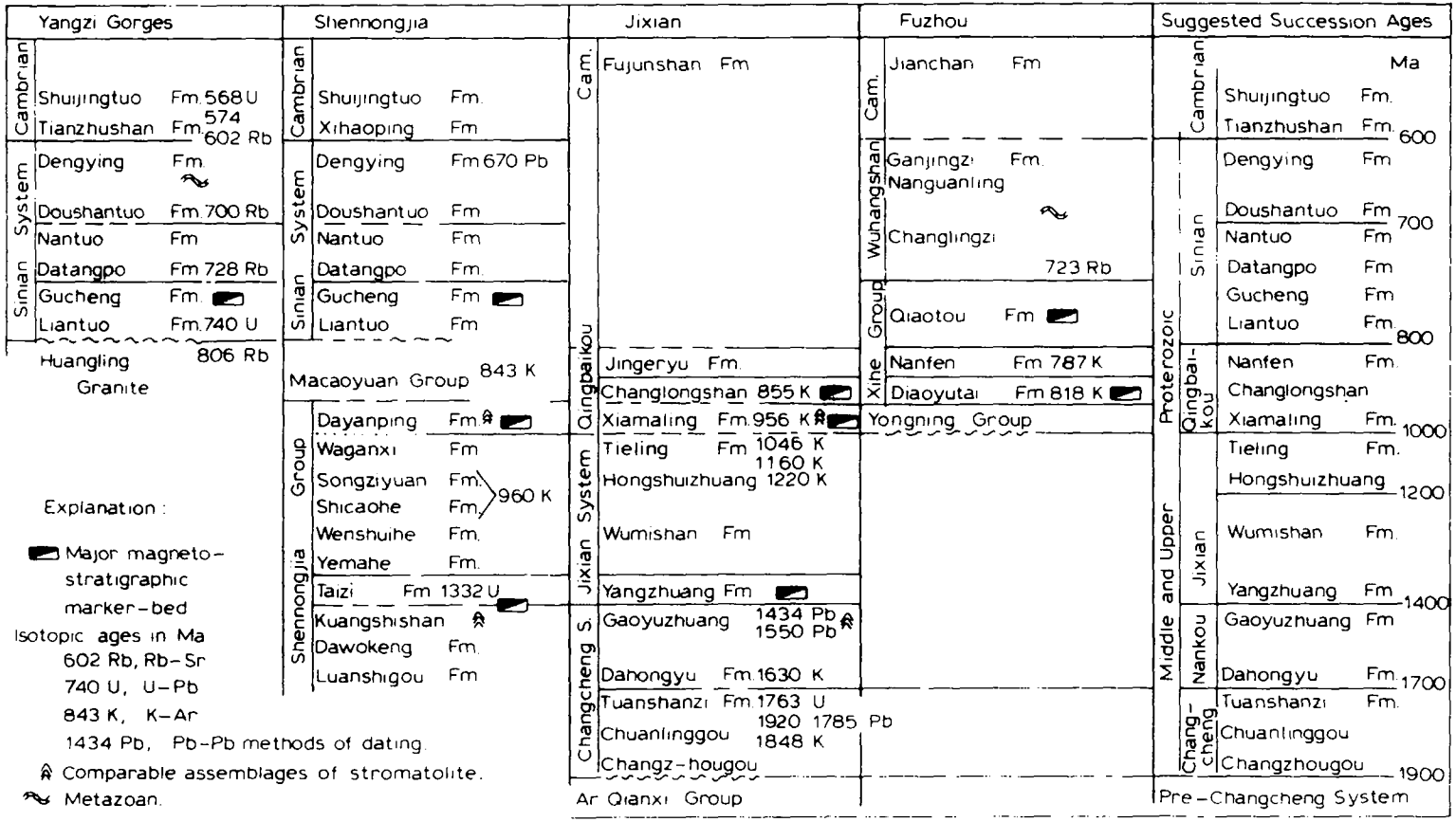
EVOLUTION OF THE TWO BLOCKS IN EASTERN CHINA

Palaeomagnetism has emerged as an important tool in tectonic studies, particularly those concerned with ancient plate tectonics. It can provide information relevant to the positions of continental blocks and palaeogeographic reconstructions (Morel and Irving, 1978; Habicht, 1979; Piper, 1982).

The following is a discussion of the tectonic evolution of the North China and Yangzi blocks during the period from 1700 Ma to 600 Ma in the light of magnetostratigraphic correlations discussed above. Assuming an axial geocentric dipole in the geologic past then: (1) during the same period of geological history the palaeopole position of continents or tectonic blocks

TABLE II

Correlation of middle and upper Proterozoic in China based on magnetostratigraphic marker-beds and other data



within continents must be the same; APW paths of continents or blocks reflect their movements relative to the palaeopole; (2) the palaeolatitude of a sampling site in a block shows the latitude of the continent or block at that time; and (3) based on the present geographic distribution, the assumption of minimum relative continental movement (Morel and Irving, 1978), is used to reconstruct the palaeogeographic position and directions of movement of continental blocks.

Using these assumptions a palaeogeographic history for the North China and Yangzi blocks has been attempted in Fig. 3. The longitudes of the blocks in the reconstructions are uncertain and are meant only to portray relative distances.

(1) Dahongyu—Luanshigou Formations (c. 1700 Ma). In this period (Fig. 3a) the two blocks in eastern China are considered to have been somewhat separated, to have lain in middle latitudes and to have been arranged nearly parallel to the longitude.

(2) Gaoyuzhuang, Dawokeng and Kuongshishan Formations to the Wumishan, Yemahe, Wenshuihe and Shicaohe Formations (Fig. 3b, c and d). In this period the two blocks were close together with no large rotational

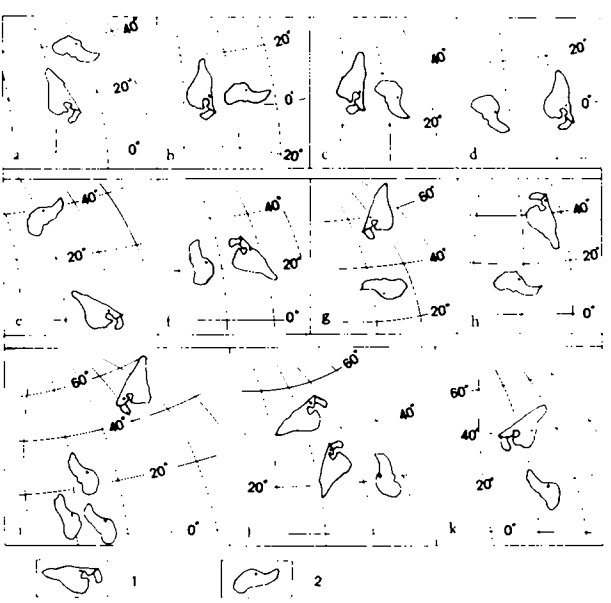


Fig. 3. Palaeogeographic distribution scheme of North China Block and Yangzi Block from c. 1700—600 Ma ago.

1. North China Block; 2. Yangzi Block.

1. Dahongyu—Luanshigou stage, b. Gaoyuzhuang—Kuangshishan stage, c. Yangzhuang—Taizi stage, d. Wumishan—Shicaohe stage, e. Hongshuizhuang—Songziyan stage, f. Xiamaling—Dayanpin stage, g. Jingeryu—Macaoyan stage, h. Qiaotou—Gucheng stage, i. Changlingzi—Datangpo, Nantuo and Doushantuo stage, j. Ganjingzi and Nanguanling—Dengying stage, k. Lower Cambrian Xingji—Shuijintuo stage.

or drift movements. They lay within 0—30° N in what was probably a tropical to subtropical environment. This milieu was suitable for development of huge thicknesses of stromatolitic carbonates (mainly dolostones) in the Yanshan rift zone of the North China block (Chen Jinbiao, 1983) and along a passive continental margin on the northern edge of the Yangzi block.

(3) Hongshuizhuang and Songziyan Formations to the Changlingzi, Datangpo, Nantuo and Doushantuo Formations (Fig. 3e—i). During the period of deposition of the Jingyeru Formation and Macaoyan Group to the time of deposition of the Changlingzi—Nantuo and Doushantuo Formations the North China block drifted into higher latitudes up to 62° N but the Yangzi block was for the most part in latitudes between 0 and 20° N. This is considered to be the period of greatest separation of the two blocks during the middle and upper Proterozoic. Such significant differences in palaeolatitude of the two blocks should be reflected in the sedimentary facies produced. As an example, the Jingeryu Formation and Nanfen Formation consist of a relatively thin but widely distributed succession of carbonates and shales. These lithofacies are very widespread and lack significant amounts of organic remains in carbonates. They are interpreted as open sea sediments, deposited in deeper water under relatively cool climatic conditions. The Macaoyan Group, on the other hand, consists mostly of thick carbonate breccias overlain by dolomitic stromatolite reefs. These deposits are considered to represent warm climatic conditions.

During deposition of the Changlingzi, Datangpo, Nantuo and Doushantuo Formations, the North China block was located at about 55° N and the Yangzi block in low latitudes between 0 and 25° N. The Changlingzi Formation is mostly sandy and argillaceous sediments deposited in subtidal and lower energy environments; it contains worm traces and Acanthophytae, the Datangpo Formation of the Yangzi block is a relatively thin manganiferous shale and dolostone succession thought to reflect warmer climates. During deposition of the Nantuo Formation, however, there was a dramatic cooling of climate in low latitudes to form extensive tillites. Several sedimentary cycles from till to outwash deposits were formed in the Shennongjia area.

This period is characterised by significant tectonic movements leading to many breaks in sedimentation and the development of several unconformities. Examples of these movements include the Qinyu Uplift in North China and the Wuling orogeny in South China.

(4) During the stage from the Dengyioing, Nanguanling and Ganjinzi Formations up to the Early Cambrian (Fig. 3j and k) the North China and Yangzi blocks were in closer proximity. Thick carbonates were deposited during this period and the crust was generally more stable.

CONCLUSIONS

In this paper we have used palaeomagnetic data to facilitate middle and upper Proterozoic stratigraphic correlations and relationships between North

China and Yangzi blocks in eastern China. These data appear to be in fair accordance with geological evidence. It should be noted, however, that the data and interpretations are of a preliminary nature and will no doubt change as the data base is expanded and improved. They do suggest, however, that palaeomagnetic and magnetostratigraphic data have great potential for resolving problems in Precambrian stratigraphy and tectonics.

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